USER GUIDE
SMART INFO KIOSK

TEAM 8: E/21/362, E/21/096, E/21/106, E/21/453, E/21/067

TEAM 15: E/21/363, E/21/416, E/21/372, E/21/054

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1.Introduction

The **Smart Info Kiosk** is a key component of the *Peraverse Crowd Management System*, designed to support visitors throughout the exhibition

The kiosk is fully interactive and helps visitors navigate the exhibition halls, discover exhibits, check event schedules, and receive real-time updates. It is designed to be simple, clear, and accessible for users of all age groups, including children, adults, and non-technical visitors.

The Smart Info Kiosk includes several pages and features:

- **Home Page:** Main navigation hub to access all features.
- **About Page:** Overview of the exhibition, its theme, purpose, and background.
- **Events Page:** List of workshops, demonstrations, and scheduled activities happening at the exhibition.
- **Exhibits Page:** A searchable and filterable list of all exhibits with detailed descriptions and locations.
- **Map Page:** A visual map to guide visitors through different exhibition areas.
- **Heat Map Page:** Shows real-time visitor density to help users avoid crowded zones.
- **Notification Page:** Important alerts, announcements, and live updates.
- **Feedback:** Allows visitors to share their thoughts, suggestions, and experience.
- **Chatbot:** A simple virtual assistant that answers common questions and helps users find information quickly.

The Smart Info Kiosk ensures that **every visitor can explore the exhibition comfortably**, even if they do not have access to the internet or are unfamiliar with technology.

2.System Requirements

The Smart Info Kiosk is designed to run on standard exhibition devices such as PCs, touch screens, or laptops. The system does not require high-end hardware but must meet the following minimum requirements to function smoothly.

1. Hardware Requirements

- Processor: Dual-Core CPU (Intel i3 or equivalent)
- RAM: 4 GB
- Storage: 2 GB free space
- Display: 18 inches or larger
- Input Device: Mouse or touchscreen
- Network: Need any internet connection

2. Software Requirements

- Operating System - Windows 10 / Windows 11
- Browser - The kiosk runs as a web application. Any browser is suitable (Google Chrome, Microsoft Edge
- Required Software – React, Node.js, PostgreSQL**.**

3.Installation & Setup Guide

- Install Required Software
- Make sure the following are installed on your computer:
  - Node.js (latest LTS version)
  - React (installed automatically when running the frontend)
  - PostgreSQL
    1. Create your own PostgreSQL user account
    2. Keep your username, password, and port ready
- Download the Project Repository
- Go to the provided GitHub link.
- Download or clone the Kiosk repository to your computer.

1. Set Up Databases & Install Dependencies Automatically

- A setup script is included inside the repository.
  1. Open the **KIOSK** folder.
  2. Locate the file: **setup_all.bat**
  3. Run the file.

2. inside every backend file, find the .js file or .env file and change the database password in to your postgres user password.

3. Run All Backends Automatically

- Inside the KIOSK folder, you will find another script: **start_backends.bat**
- This file will: Start all backend services, Load environment variables, connect to PostgreSQL using the credentials you entered
- To use it:

1. Go to **KIOSK** folder
2. Open the start_backends.bat file and enter your **PostgreSQL username** and **password** in the provided space
3. Run the file **start_backends.bat**
4. All backend servers will start automatically in separate terminal windows

4. Installation and Start Chatbot (Windows PowerShell)

1. Open PowerShell and change to the chatbot-server folder: Set-Location **'KIOSK\chatbot-server'**
2. Install dependencies: **npm install**
3. Start the server: **node knowledge-base-api.js**

5. Run the Frontend

1. Navigate to the frontend folder (**KIOSK\frontend**).
2. Run the frontend: **npm install** and **npm run dev**

The kiosk interface will now be available at the URL displayed by Vite (usually http://localhost:5173).

4.How to Use the Kiosk Pages

1. Home Page

The Home Page is the first screen that appears when users interact with the Smart Kiosk. It is designed to be simple, clean, and easy to use.



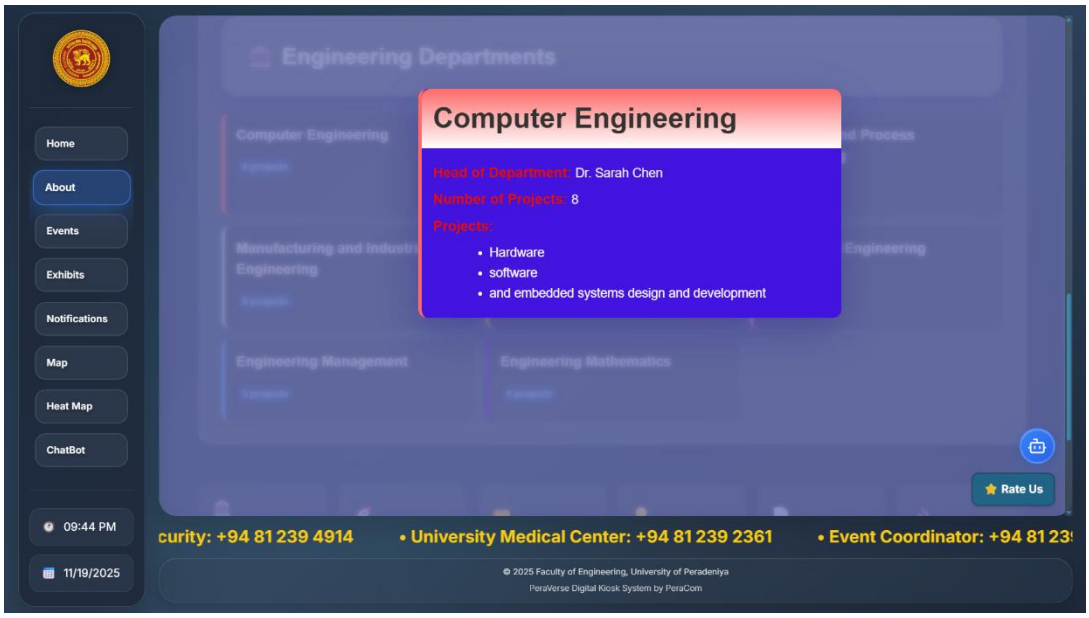
2. **About Page**

Consist of the details about the event and the general information about the faculty and the university.

In the main Navigation Panel, there is a button called About. Click it all the information will appear.



Click the department card inside the Engineering Departments category: it will appear a separate description box for each department which consist of basic information

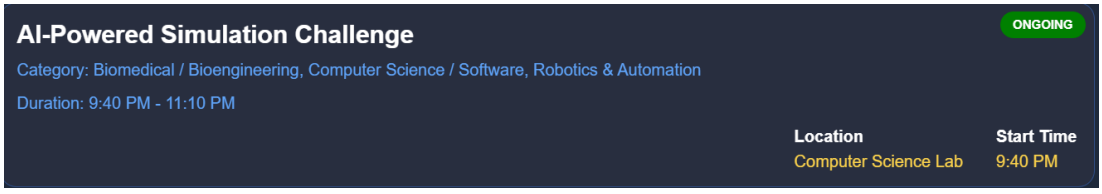


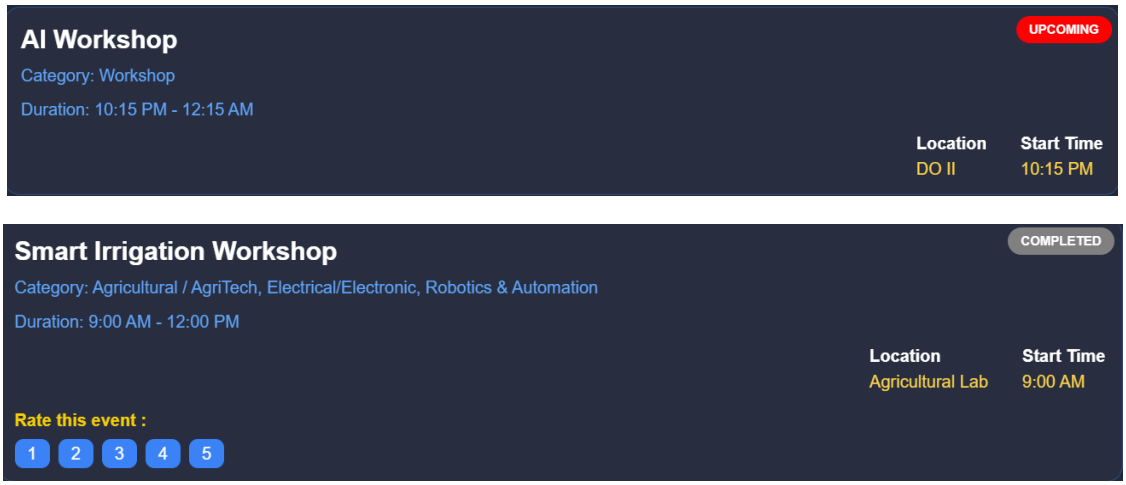
3. Events Page

Key features:

1. Viewing Event Details

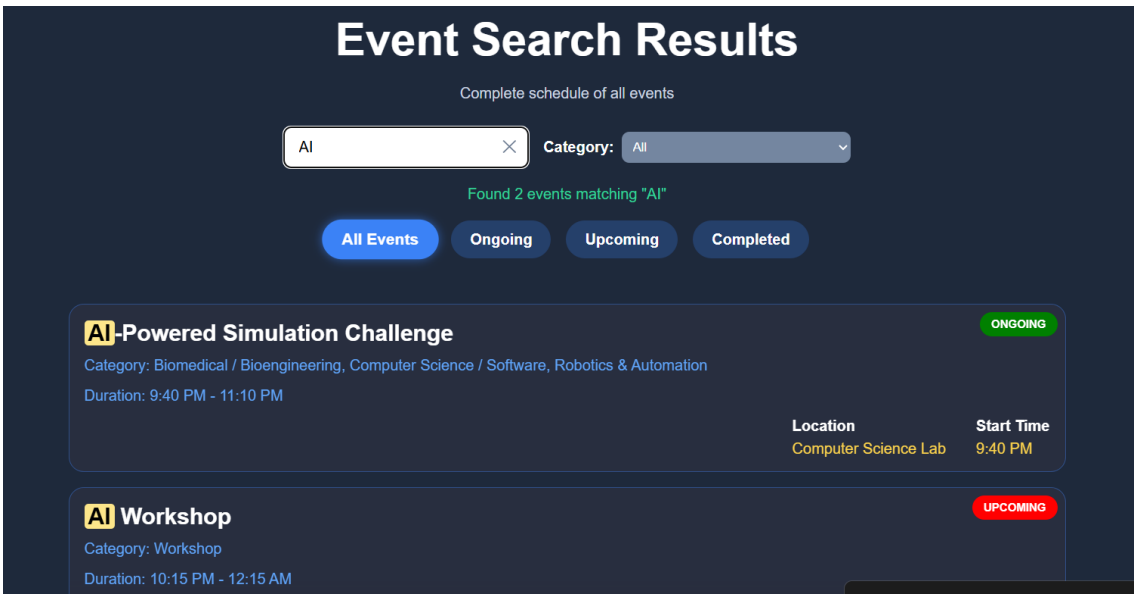
- Each event card shows:
    - **Event Title**
    - **Category**
    - **Duration** (Start – End Time)
    - **Location**
    - **Status Badge** (Ongoing / Upcoming / Completed)
- **Event Status Colors:**
    - **Ongoing:** Green
    - **Upcoming:** Red
    - **Completed:** Gray





2. Searching Events

- Use the **search box** to find events by **title, location, start, or end time**.
- The kiosk will highlight all text that matches the search keyword in **yellow**.
- A message displays the number of matching events, e.g., Found 2 events matching "AI".
- Click the ✕ **button** to clear the search query.

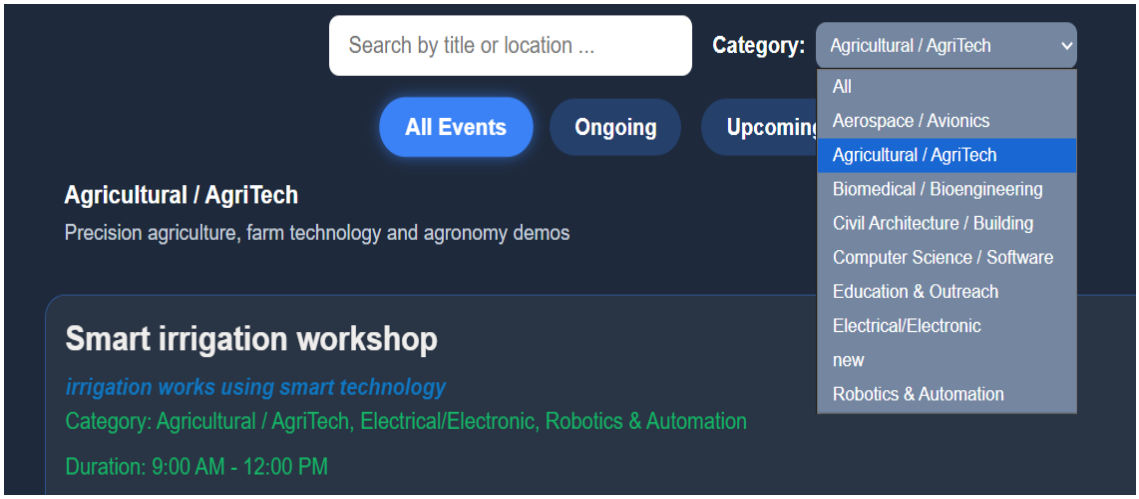


3. Filtering Events

<u>Category Filter</u>

- A simple dropdown list that displays all available event categories.

- Categories are fetched directly from the system database.
- Each category shows both its name and description.
- Once a category is selected, only the events belonging to that category will be displayed.



How to use:

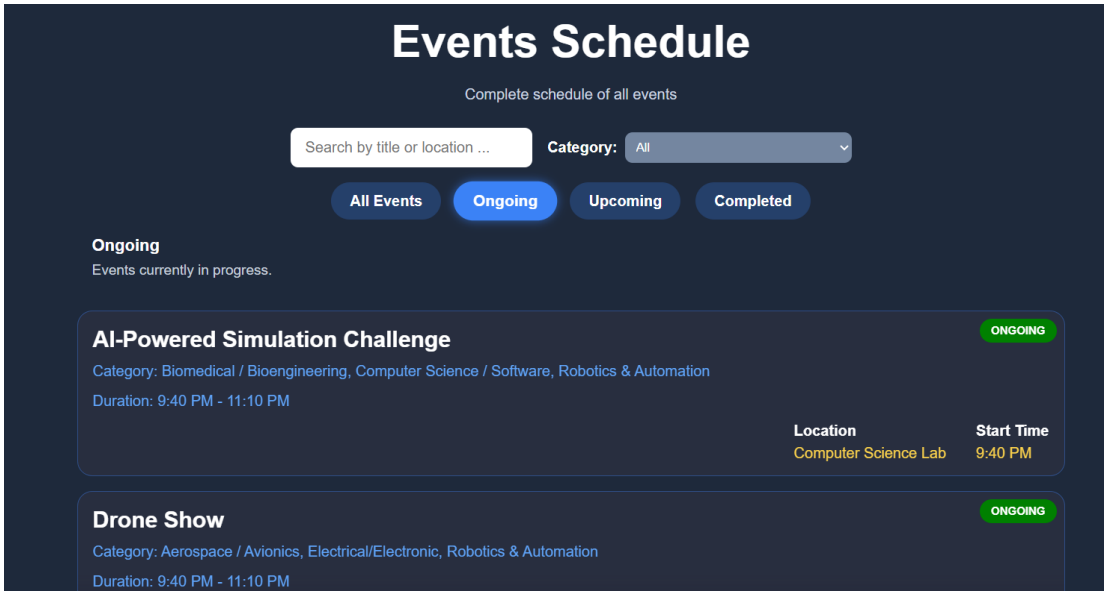
- Open the Event page from the Home Page.
- Tap the Category Dropdown.
- Select the category you are interested in (e.g., *Robotics & Automation*).
- The page will automatically show only the events related to that category.
- Scroll through the list to explore each event.

Why this feature useful:

- Saves time by narrowing down events based on your interests.
- Helps visitors quickly find relevant exhibits or talks.
- Avoids the need to scroll through all events manually.

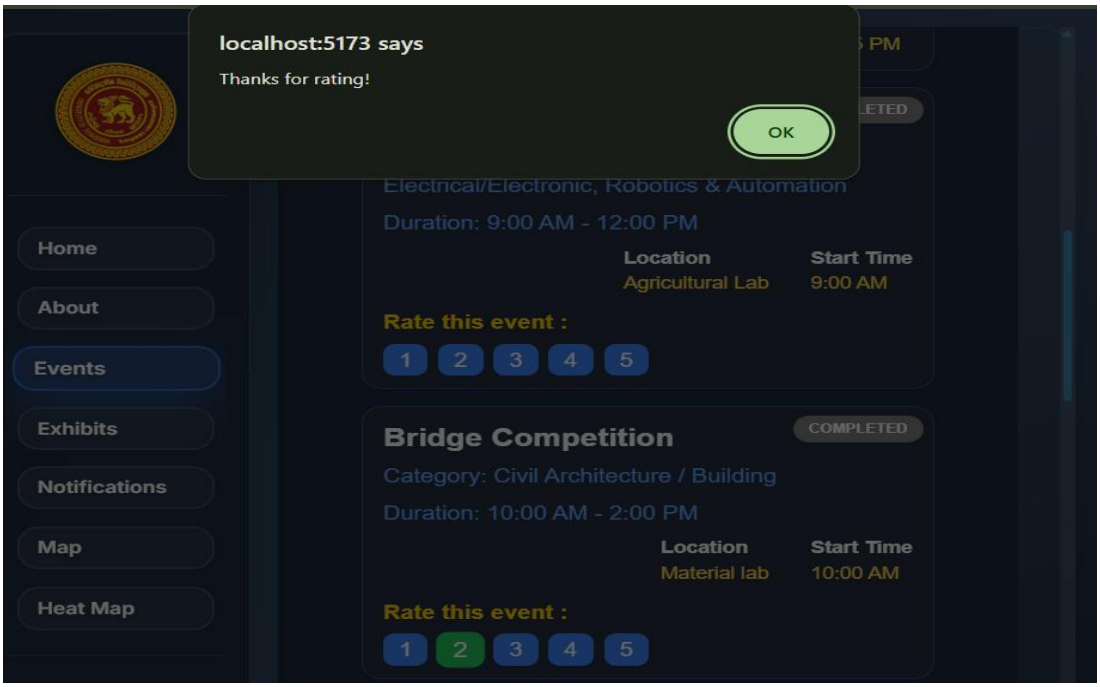
Status Filter

- Choose from **All, Ongoing, Upcoming, Completed** buttons to filter events.



4.Rating Completed Events

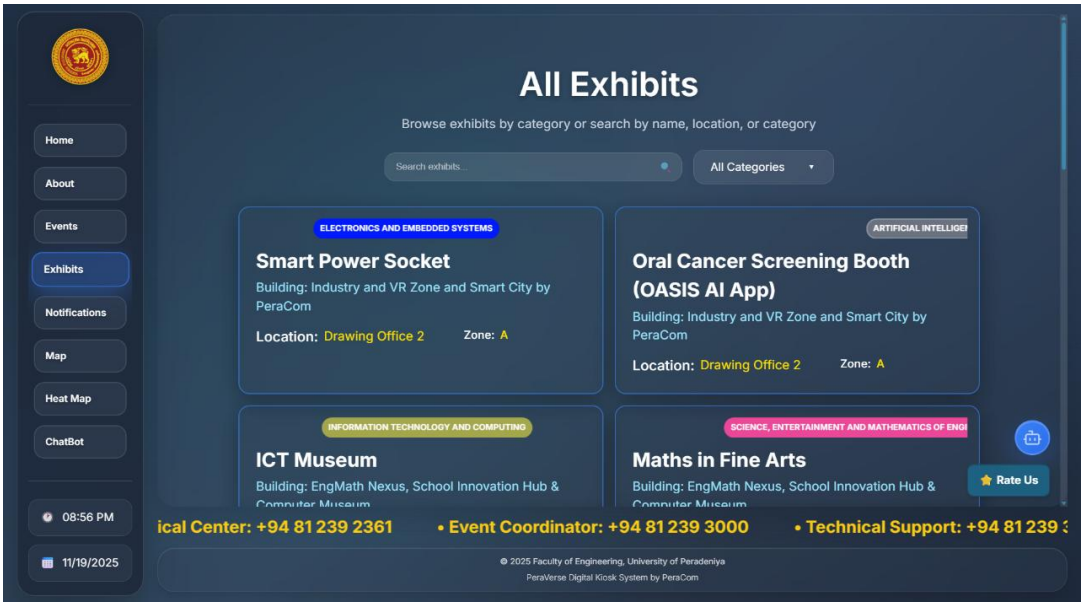
- Completed events display a **rating section**.
- Click a number (1-5) to rate the event.
- A small popup confirms submission with the message: **"Thanks for rating!"**



4. **Exhibits Page**

Consist of all exhibits in the exhibition. Display the exhibit name location, Tags that relate to the exhibit. All data saved in a PostgreSQL database.

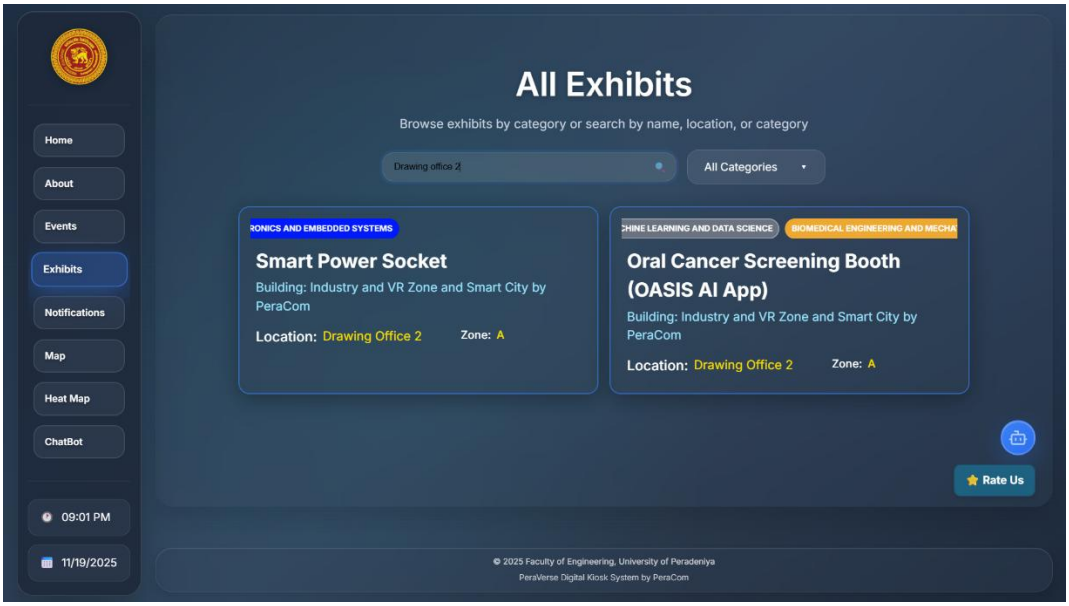
In the main Navigation Panel, there is a button called Exhibits. Click it all the exhibits will appear.



1.Filter by Tags: There is a dropdown list in right up corner you can filter Exhibits by their Tags.

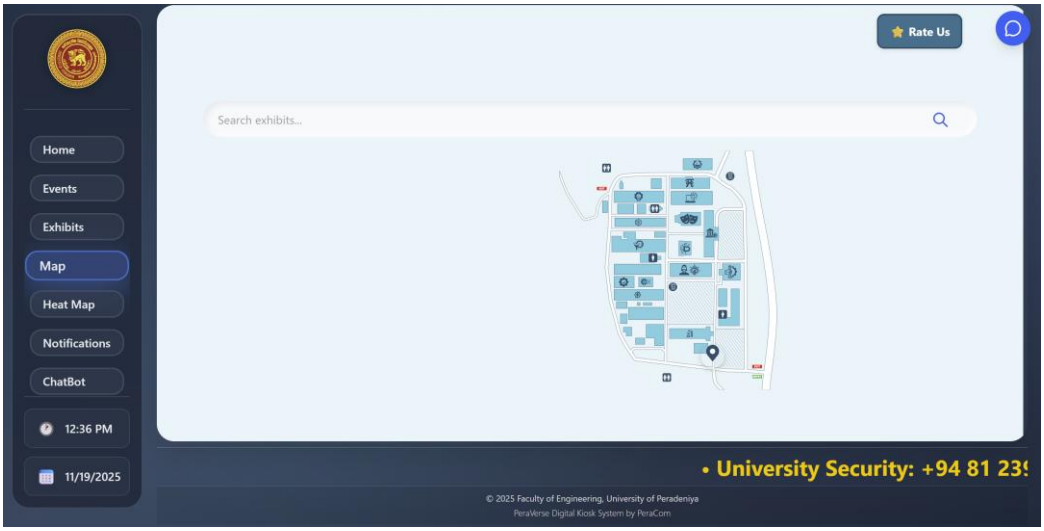


2.Search bar: You can search any exhibit by the Name, Location or Tags



5. **Map Page**

the map feature helps users easily locate buildings and exhibits inside the kiosk system.



1. Viewing buildings on the map.

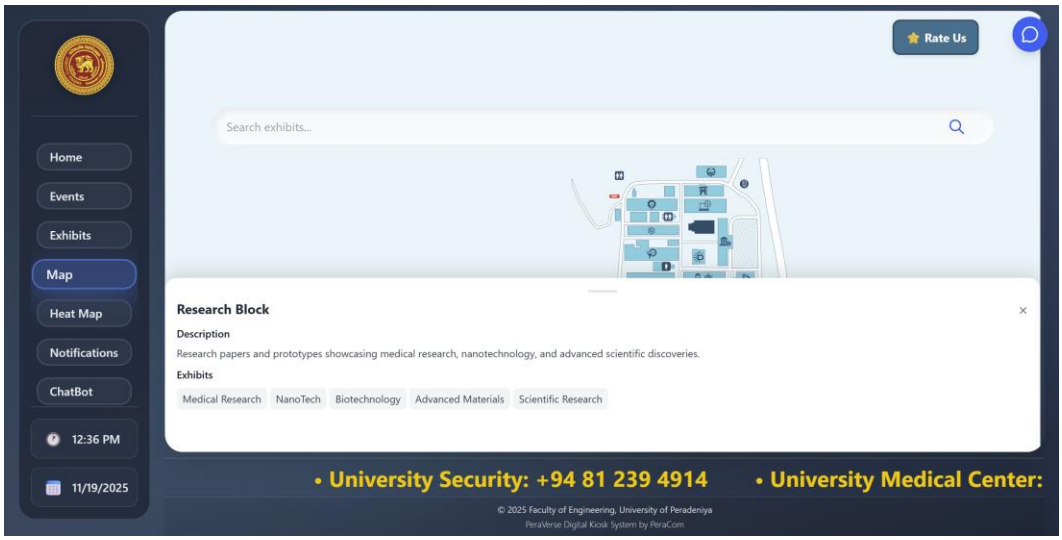
   When the Map tab is opened, the full layout of the area is displayed.

   Each building appears on the map and can be selected to view additional information.

   After a building is selected, the system displays:

   - The building name
   - The exhibits available inside that building

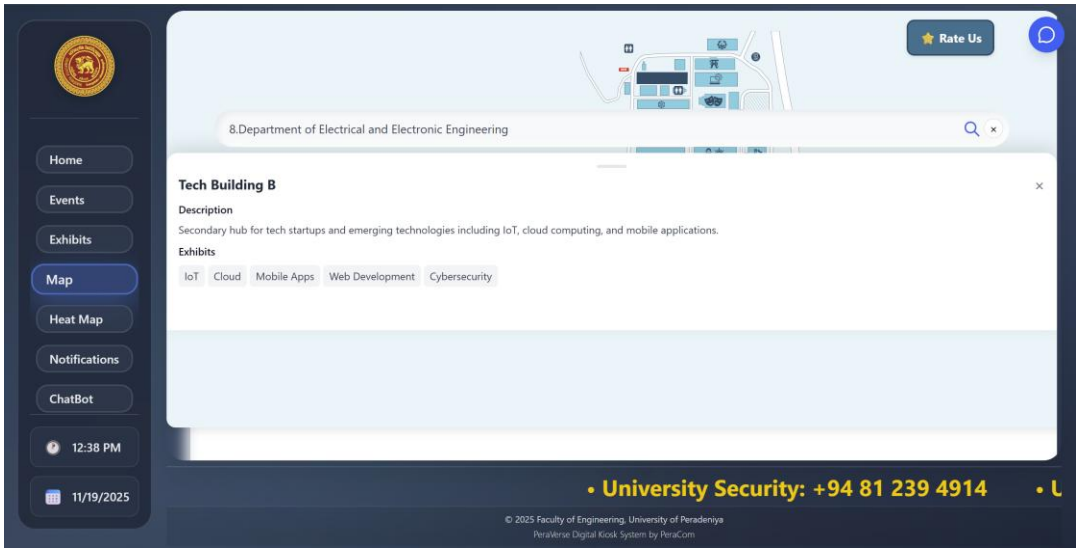
This allows quick understanding of the exhibitions located in each place.



2. Using the search bar.
   A search bar is available at the top of the Map screen.
   It can be used to find **buildings** or **exhibits**.
   - Begin typing the name of a building or exhibit.
   - A dropdown list appears with all matching results.
   - Selecting an item from the list will proceed to the next step.



3. Highlighting locations on the map.
   After a building or exhibit is selected from the search results:
   - The corresponding building on the map becomes **highlighted** or **colored**.
   - This visual indication helps in easily identifying the building's exact location on the map.



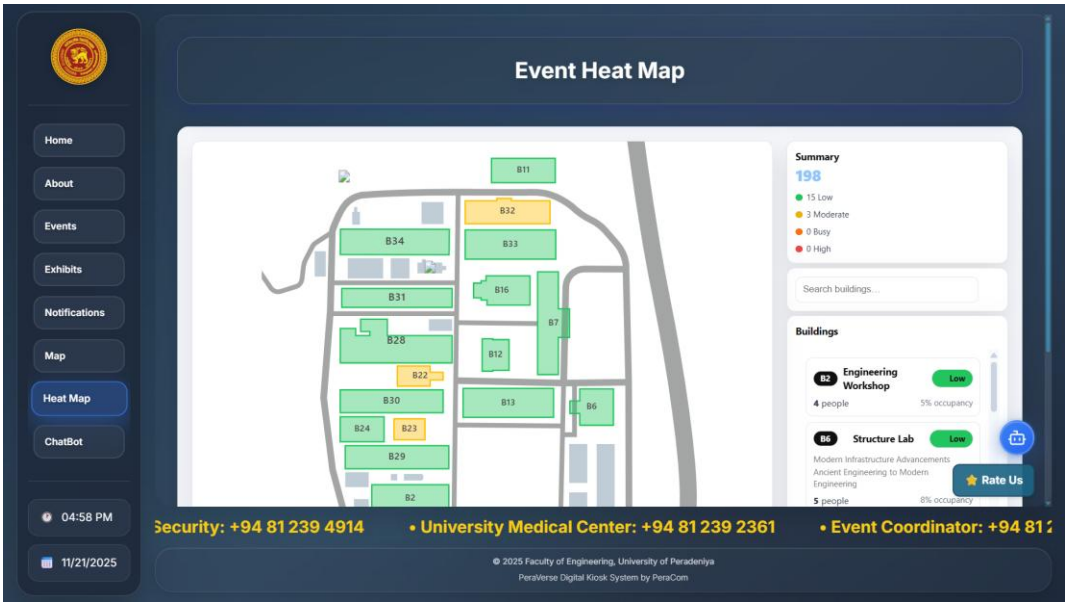
6. **Heat Map Page**

Show real time crowd density of the exhibition.

Colored buildings: It uses the exhibition map and use three colors to show the density

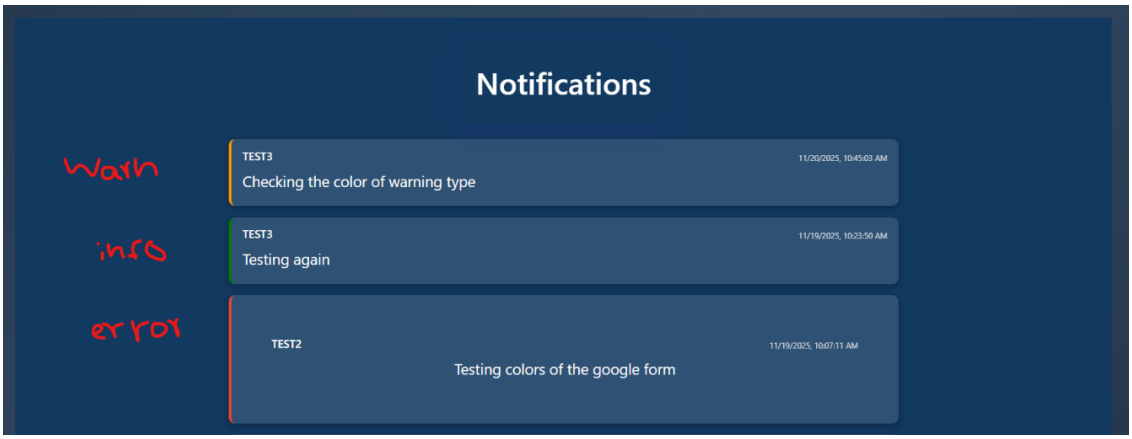
- Green: Low
- Orange: Moderate
- Red: Crowded

Details Bar: Shows the count of the people in each building and what is the maximum capacity of that building.



7. Notification Page

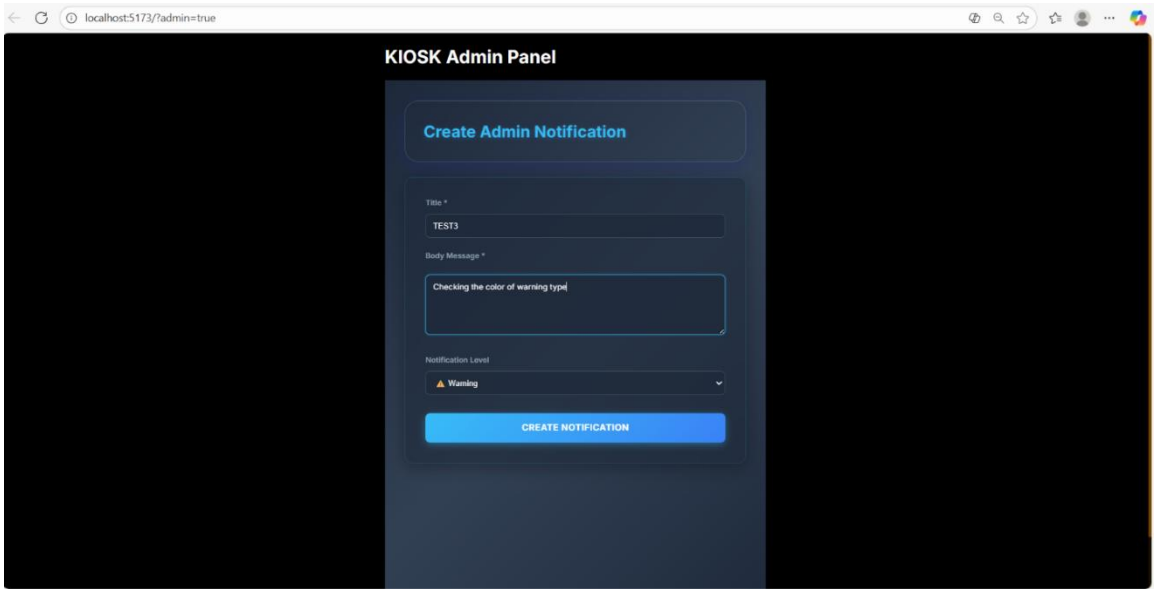
The page automatically: Refreshes every 5 seconds to show new notifications, Updates countdown timers every second, Removes expired notifications after events start, No manual refresh needed - everything updates automatically.



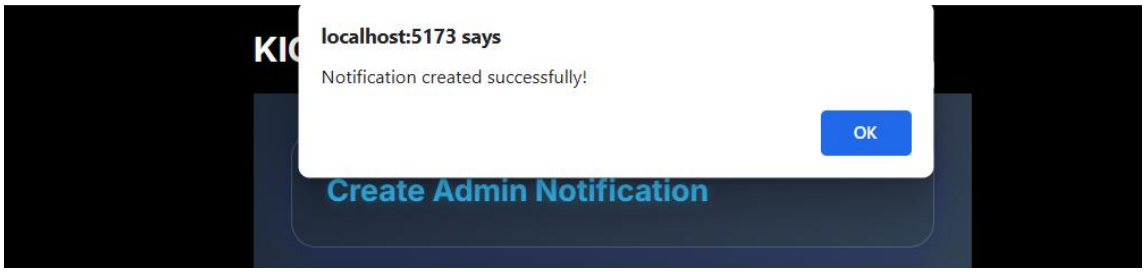
Accessing the Admin Panel

Step 1: Ensure all servers are running (notifications backend, events backend, and frontend).

Step 2: Open your web browser and navigate to: http://localhost:5173/?admin=true



You can create a Custom Notifications, after creating

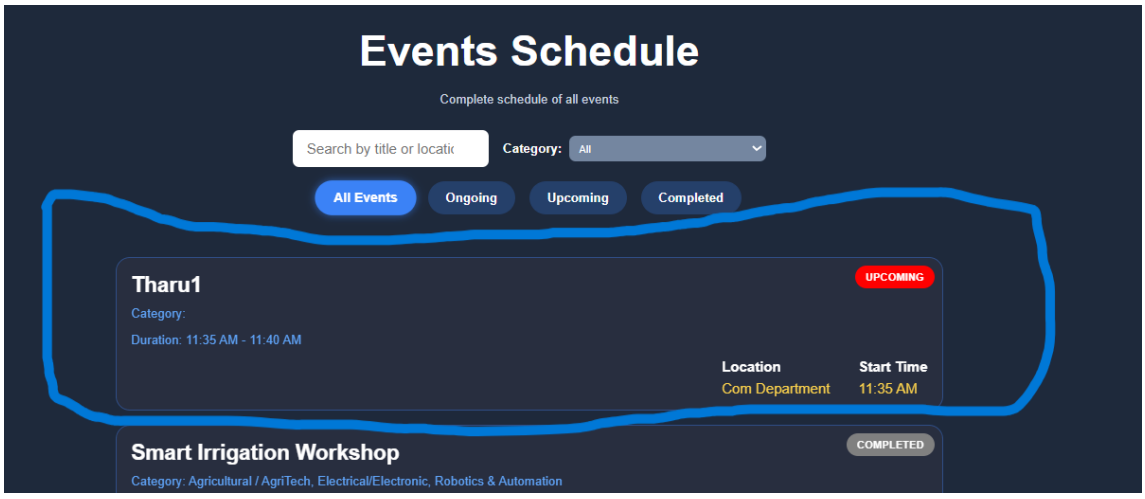


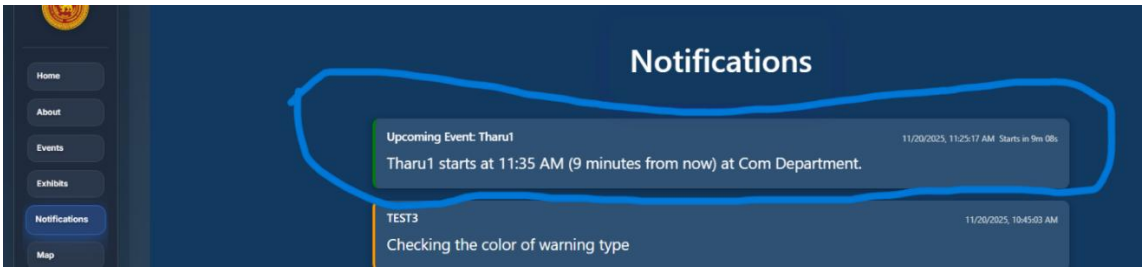
The form clears automatically, The notification is now live on the Notifications page

Automatic event notifications - Generated from Events Schedule for events starting within 30 minutes

In psql command prompt (to check),

```
       | Biomedical Lab       | 2025-09-23 13:00:00        | 2025-09-23 16:00:00    Cancel request sent
event_db=# INSERT INTO events(event_title,location,start_time,end_time,category_id)
event_db-# VALUES ('Tharu1','Com Department',NOW() + INTERVAL '10 minutes',NOW() + INTERVAL '15 minutes',6);
INSERT 0 1
event_db=#
```





Notification will automatically delete when the event starting time.

7. Feedback Popup

The Feedback Popup is a small tool added to the kiosk system to let users quickly give a 1-5 stars rating and write optional comments. This helps the admin understand user experience and improve services.

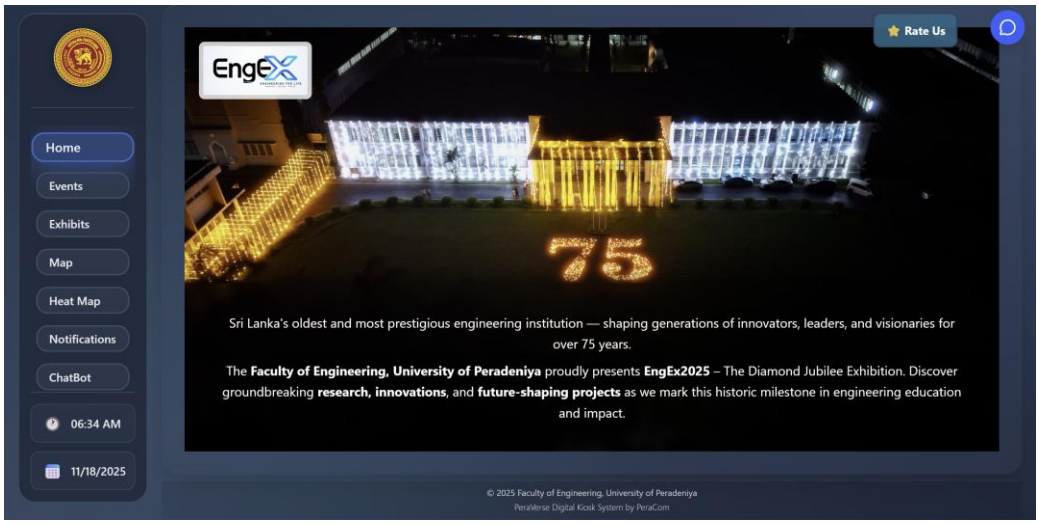
All feedbacks are automatically saved in two places.

1) PostgreSQL Database – main data storage
2) CSV (Excel) file – automatic backup file (feedbacks.csv)

The feature is fully anonymous and simple to use.

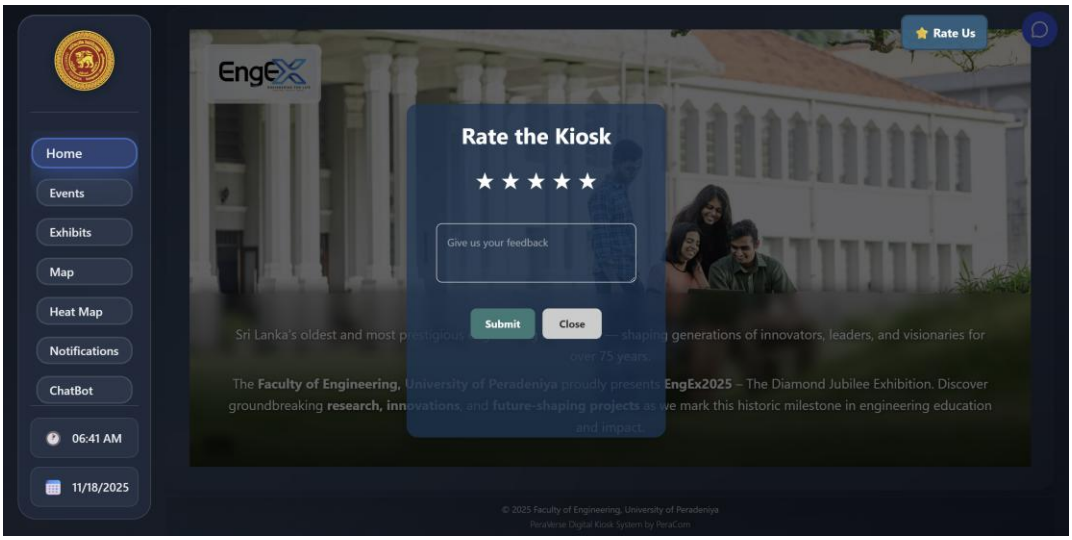
Users will see a floating button on the screen: Simple to use

⭐ Rate Us

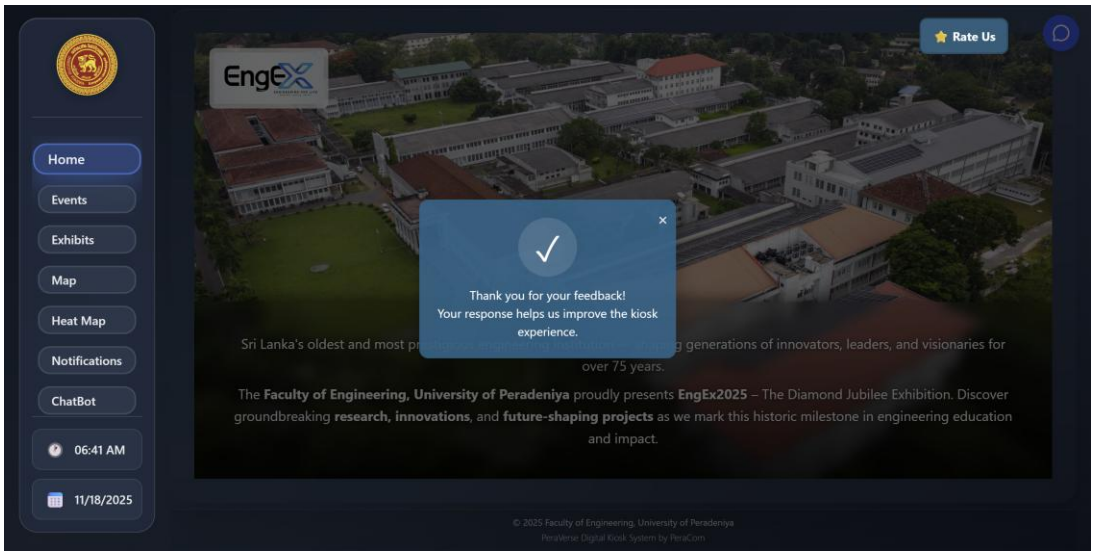


When they click it:

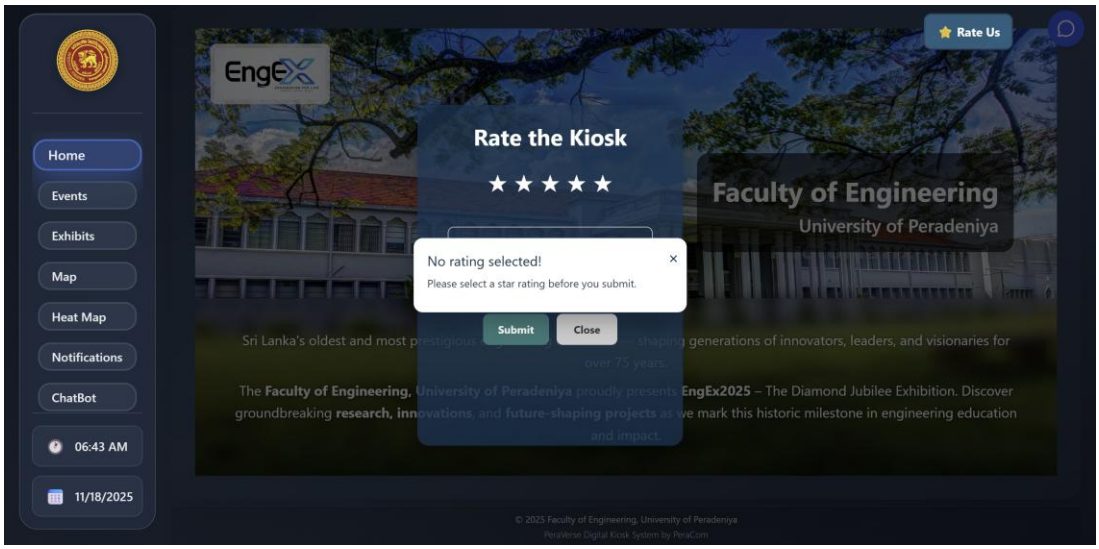
- A popup opens
- They can choose stars
- They can type a message (optional)



- When they click Submit, a thank you message is shown



- If they forget to select a star rating, a message appears asking them to choose one.



8. **Chatbot**

1.The frontend sends a query to POST /api/knowledge-base/query.

2.The Server: Validates input (query required), Extracts keywords (splits, lowercases, filters 3.Short words) Checks the in-memory cache, Queries Supabase if not chached, Returns structured JSON results.

4.A simple health endpoint provides availability checks.

5.Performance & Runtime Behavior

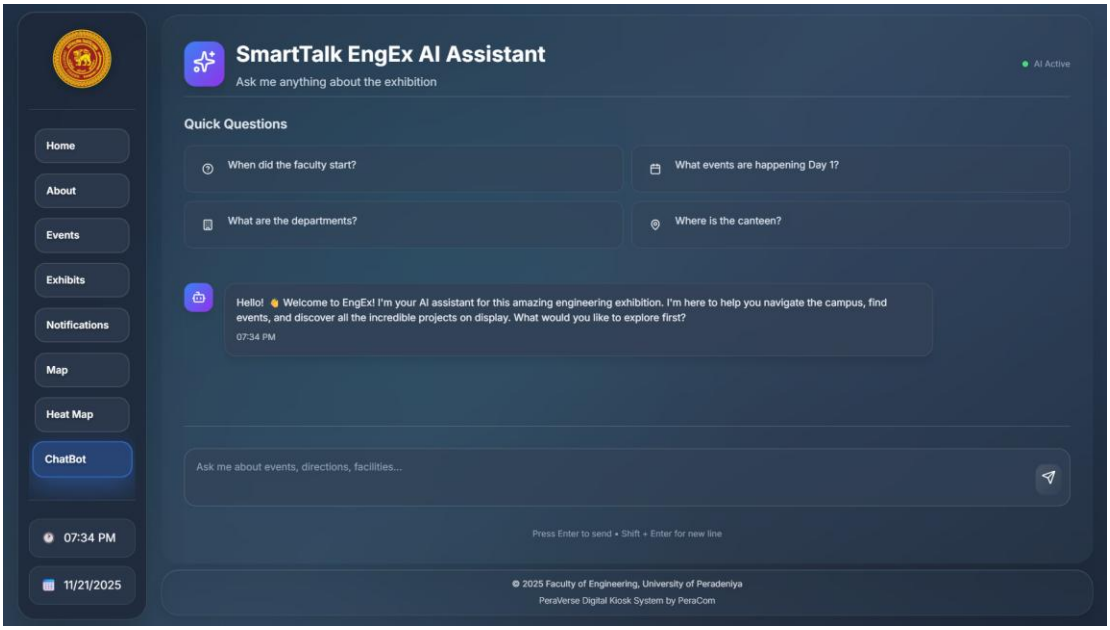
Response Caching: Short-term in-memory cache reduces repeated DB hits. Cache resides in server process memory; restart clears cache.

Rate Limiting: Requests are rate-limited per IP to protect against overload (client receives a 429-like response message when limit exceeded).

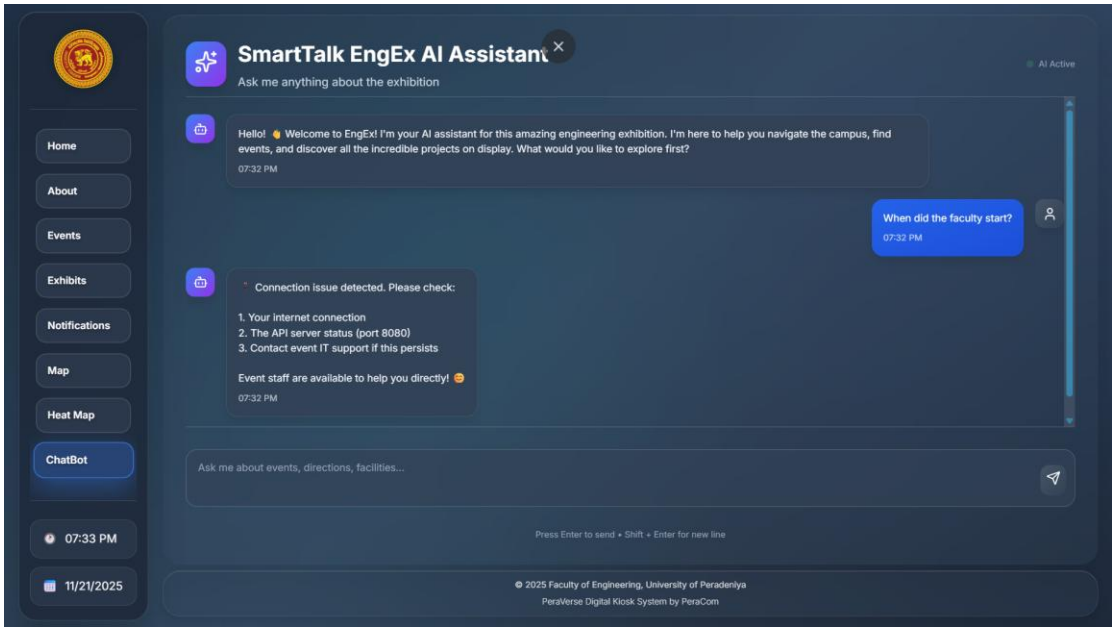
Health Checks: Use /health in monitoring to detect service outages.

6.How to use:

the search information is only limited to the event details. Users can search the information related to the event.
when search for locations, automatically connect to the map.



7.when there is an error it shows:



5.Troubleshooting Guide

General Issues

App not loading / blank screen: Ensure backend is running (run-backends.bat) and frontend started (npm run dev), refresh browser, use full-screen (F11).

Buttons/pages not responsive: Reload page, clear cache, restart browser, rebuild frontend (npm run build).

Backend & Database

Backend fails to start: Ensure PostgreSQL service is running, username and password are correct, check that the same credentials are properly set in all backend configuration files, database exists, close port conflicts, run npm install.

Exhibits page empty / filters not working: Run setup-all.bat, ensure backend running, PostgreSQL credentials are correct in both database and backend configs, refresh page.

Database scripts fail: Use a PostgreSQL user with createdb permission, verify username/password, run .bat as Administrator.

Hardware / Device

Kiosk not turning on: Check power cable, outlet, and PC switch.

Display shows “No Signal”: Check cables, restart PC, verify monitor input.

Touchscreen not responding: Recalibrate, restart device, use supported browser.

Feature-Specific

Chatbot not responding: Ensure backend running, check network.

Map / Heatmap not showing: Backend not running, refresh page, check API connection.

Support

If problems persist, ensure credentials are correct both in PostgreSQL and backend configs, note the error, take a screenshot, and contact the technical team.

6.Contact & Support

If you experience any issues with the Smart Info Kiosk that cannot be resolved using the Troubleshooting Guide, please use the following contact information to get assistance.

Technical Support

For problems related to installation, backend services, database connection, or system errors.

- **Contact Person:** E/21/362
- **Email:** e21362@eng.pdn.ac.lkp

7.References

1.Google Developer Documentation Style Guide: https://developers.google.com/style Google for Developers+1

2.Git User Manual: https://git-scm.com/docs/user-manual "How to Create a User Manual"

3.Here's a link to the official GitHub (Docs) user-manual/reference page you can use: Legacy Manual – GitHub Training Team (PDF)